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Algebraic Properties

$$y = Q + \hat{e}$$

$$\hat{\beta} = (X'X)^{-1}X'y$$

$$\hat{Q} = X\hat{\beta}$$

$$y' - \hat{e}' = 0$$

chapter 4

A 4.1 $(y_i, x_i) \sim i.i.d.$

A 4.2 Linear model : $y_i = x_i' \beta + e_i$

$$E[e_i | x_i] = 0$$

$$E[y_i^2] < \infty \quad E[\|x_i\|^2] < \infty$$

$$Q_{xx} = E[x_i x_i'] > 0$$

A. 4.3 $E[e_i^2 | x_i] = \sigma^2(x_i) = \sigma_c^2$

no serial correlation
homoskedasticity assumption.

$e_i, i=1, \dots, n.$

e
 $n \times 1$

$$E[ee' | X] = \begin{bmatrix} E(e_1^2 | X) & E(e_1 e_2 | X) & \dots & E(e_1 e_n | X) \\ \vdots & & & \\ E(e_n e_1 | X) & \dots & \dots & E(e_n^2 | X) \end{bmatrix}$$

$\underbrace{\quad}_{n \times 1 \quad 1 \times n}$
 $\underbrace{\quad}_{n \times n}$

$$= D$$

no serial correlation : $E(e_i e_j | X) = 0, i \neq j$
 off diag of D are 0's.

homoskedasticity : $E(e_i^2 | X) = \sigma^2$
 constant variance assumption,

$$E(e | X) = 0 \quad \left(\begin{array}{l} E(e_1 | X) \\ E(e_2 | X) \\ \vdots \\ E(e_n | X) \end{array} \right) = 0 = E(e_i | X_1, \dots, X_n)$$

$$\text{By iid, } E(e_i | X) = E(e_i | X_i)$$

$$\text{By iid, } E(e_i | X) = 0 \quad \text{same as} \quad E(e_i | X_i) = 0$$

$$E(e_i | X_i) = 0 \quad \Rightarrow \quad E(X_i' e_i) = 0$$

—
conditional moment.

—
orthogonality
condition.

$$\hat{\beta} = (X'X)^{-1} X'y$$

DGP: $y = X\beta^0 + e$

$$\begin{aligned}\hat{\beta} &= (X'X)^{-1} X' (X\beta^0 + e) \\ &= \cancel{(X'X)^{-1} X' X} \beta^0 + (X'X)^{-1} X'e\end{aligned}$$

$$\hat{\beta} = \beta^0 + (X'X)^{-1} X'e$$

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$$\begin{aligned}E[\hat{\beta} | X] &= \beta^0 + E[(X'X)^{-1} X'e | X] \\ &= \beta^0 + (X'X)^{-1} X' \underbrace{E[e | X]}_{=0}\end{aligned}$$

$$\begin{aligned}E[\hat{\beta}] &= E_X [E(\hat{\beta} | X)] \\ &= \beta^0\end{aligned}$$

$$\begin{aligned}V(\hat{\beta} | X) &= E[(\hat{\beta} - E(\hat{\beta} | X)) (\hat{\beta} - E(\hat{\beta} | X))' | X] \\ &\quad \underbrace{K \times K} \quad \underbrace{K \times 1} \quad \underbrace{1 \times K} \\ &= E[(\hat{\beta} - \beta^0 | X) (\hat{\beta} - \beta^0 | X)' | X] \\ &= E[(X'X)^{-1} X'e e' X (X'X)^{-1} | X] \\ &= (X'X)^{-1} X' \underbrace{E[ee' | X]}_{=I} X (X'X)^{-1}\end{aligned}$$

$$\begin{aligned}
 &= \underbrace{(X'X)^{-1}}_{\text{bread}} X' \underbrace{D}_{\text{meat}} X \underbrace{(X'X)^{-1}}_{\text{bread}} \\
 &= V_{\hat{\beta}} \quad \text{sandwich form}
 \end{aligned}$$

special case, no serial correlation
homoskedasticity $] D = \sigma^2 I_k$

$$\begin{aligned}
 \text{Then } V_{\hat{\beta}} &= (X'X)^{-1} X' \sigma^2 I_k X (X'X)^{-1} \\
 &= \sigma^2 (X'X)^{-1}
 \end{aligned}$$

Gauss Markov Theorem

$\hat{\beta}$ is BLUE (Best linear unbiased estimator)

$$\hat{\beta} = (X'X)^{-1} X' y$$

$$\hat{\beta} = A y \quad \text{OLS: } A = (X'X)^{-1} X'$$

Any linear unbiased estimator must satisfy $A'X = I_k$.

requires homoskedasticity.

$$Y = X\beta + e$$

$$\hat{\beta} = (X'X)^{-1} X'Y$$

$$\hat{e} = Y - X\hat{\beta}$$

$$= M Y = M e$$

$$e_i \sim (0, \sigma^2)$$

$$E(e_i^2) = \sigma^2$$

$$E(e_i^2) = \sigma^2$$

$$\text{MLE} \quad \leftarrow \quad \hat{\sigma}^2 = \frac{1}{n} \sum_i \hat{e}_i^2 \quad \text{biased estimate}$$

$$\text{OLS} \quad \leftarrow \quad s^2 = \frac{1}{n-k} \sum_i \hat{e}_i^2 \quad \text{unbiased}$$

$$\hat{e} = M e$$

$$\begin{aligned} E[\hat{e} \hat{e}' | X] &= E[M e e' M' | X] \\ &= M E(e e' | X) M' \\ &= M D M' \end{aligned}$$

Remark: even if e is serially uncorrelated, $\hat{e}$ will be correlated

Prediction:

$$e_i = y_i - x_i' \hat{\beta}$$

(in sample
residuals)

$$\hat{y}_{n+1} = x_{n+1}' \hat{\beta}$$

$$\hat{e}_{n+1} = y_{n+1} - x_{n+1}' \hat{\beta}$$

$$= x_{n+1}' \beta + e_{n+1} - x_{n+1}' \hat{\beta}$$

$$= x_{n+1}' (\underbrace{\beta - \hat{\beta}}_{-1/n+1}) + e_{n+1}$$

$$E(\hat{e}_{n+1}^2) = E(e_{n+1}^2) + x_{n+1}' (X'X)^{-1} x_{n+1}$$

$$\text{since } V(\hat{\beta}) = \sigma^2 (X'X)^{-1}$$

Omitted variable bias

$$X = [X_1 \quad X_2]$$

DGP: $Y = X_1 \beta_1 + X_2 \beta_2 + e$

Regression: $Y = X_1 b_1 + u$

$$\hat{b}_1 = (X_1' X_1)^{-1} X_1' Y$$

$$= (X_1' X_1)^{-1} X_1' (X_1 \beta_1 + X_2 \beta_2 + e)$$

$$= \beta_1 + (X_1' X_1)^{-1} X_1' X_2 \beta_2 + (X_1' X_1)^{-1} X_1' e$$

$$E(\hat{b}_1 | X) = \beta_1 + (X_1' X_1)^{-1} X_1' X_2 \beta_2$$

biased except when $X_1 \perp X_2$

Inclusion of irrelevant regressor

DGP: $Y = X_1 \beta_1 + e$

Regression: $Y = X_1 b_1 + X_2 b_2 + u$

What is $\hat{b} = \begin{pmatrix} \hat{b}_1 \\ \hat{b}_2 \end{pmatrix}$?

$$E[\hat{b} | X] = \beta = \begin{bmatrix} \beta_1 \\ 0 \end{bmatrix}$$

$$\text{var}(\hat{\beta}_1 | x_1) > \text{var}(\hat{\beta} | x_1, x_2) > 0$$

Small regression $\nearrow$ bias
 $\searrow$ smaller variance.

$$\text{MSE} = \text{bias}^2 + \text{variance}.$$

Ridge Estimation

multicollinearity: K regressors are highly dependent

conditioning index, ratio $\left(\frac{\text{largest}}{\text{smallest}} \right)$ eigen.

$$X = U D V' \quad \begin{matrix} \text{matrix of singular} \\ \text{values.} \end{matrix}$$

$n \times n \quad n \times K \quad K \times K$

$$\begin{aligned} \hat{\beta}_R &= (V D^2 V')^{-1} V D U' y \\ &= (X'X + \lambda I_K)^{-1} X' y \\ &= V (D^2 + \lambda I_K)^{-1} D U' y \end{aligned}$$

$$X \hat{\beta}_R = \sum U_j \left(\frac{d_j^2}{d_j^2 + \lambda} \right) U_j' y.$$

$$X \hat{\beta}_{OLS} = \sum_{j=1}^K U_j U_j' y = U U' y. \quad \text{shrinkage.}$$

$$\hat{\beta}_2 = \begin{pmatrix} \hat{\beta}_1 \\ 0 \\ 0 \\ \vdots \\ \hat{\beta}_{k+1} \end{pmatrix}$$

$$\hat{\beta}_R = \begin{pmatrix} \hat{\beta}_1^* + \epsilon \\ \hat{\beta}_1 \\ \vdots \\ \hat{\beta}_n \end{pmatrix}$$

$$y_i = x_i \beta + e_i$$

$$e \sim N(0, \sigma^2)$$

$$\hat{\beta} = \beta + (X'X)^{-1} X'e$$

classical normal linear model

$$\hat{\beta} - \beta \sim N\left(0, \underbrace{(X'X)^{-1} X' \Omega X (X'X)^{-1}}_{M_{\hat{\beta}}}\right)$$

fully Parametric model.

Large Sample Analysis

① random sequence $b_1, \dots, b_2, \dots$

converges in probability to b if for every $\varepsilon > 0$, $\lim_{n \rightarrow \infty} P(|b_n - b| \geq \varepsilon) = 0$

p.lim $b_n = b$.

②. Mean square convergence implies convergence in probability.

$$\begin{aligned} P(|b_n - b| > \varepsilon) &= P(|b_n - b|^2 > \varepsilon^2) \\ &\leq E[(b_n - b)^2] / \varepsilon^2 \\ &= \text{Var}(b_n) / \varepsilon^2 \\ &\rightarrow 0 \quad \text{if} \quad \text{Var}(b_n) \rightarrow 0 \end{aligned}$$

an estimator $\hat{\beta}$ is consistent if $\text{Var}(\hat{\beta}) \rightarrow 0$ as n increases.

$$\begin{aligned} x_i \sim (\mu, \sigma^2) \quad \frac{1}{n} \sum x_i &= \bar{X} \\ \bar{X} &\xrightarrow{P} \mu \end{aligned} \quad \begin{aligned} \text{Var}\left(\frac{1}{n} \sum x_i\right) &= \frac{1}{n^2} \sum \sigma^2 \\ &= \frac{\sigma^2}{n} \end{aligned}$$

check $\text{Var}(\bar{X}_n) \rightarrow 0$ as $n \rightarrow \infty$

$$\text{Var}(\bar{X}) = \frac{\sigma^2}{n} \rightarrow 0$$